Software Requirements Specification (SRS) Template

Project: Student attendance management system - An app like the app we have at PESU

Version: 1.0 **Authors:** < Instructor / Student> **Date:** 29-08-2025 **Status:** Model Answer / Approved

Revision history

Version	Date	Author	Change summary	Approval
1.0	29-08-2025	Instructor	SRS with diagrams embedded	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This SRS specifies the requirements for the **Student Attendance Management System (SAMS)**. It defines functional and non-functional requirements, user interfaces, security goals. The system aims to automate attendance tracking for students, ensuring accuracy, real-time reporting.

1.2 Scope

SAMS allows faculty to record attendance digitally, students to view attendance reports. Scope includes:

- Attendance marking
- Reports and analytics

1.3 Audience

Faculty & Students

1.4 Definitions

- **SAMS** – Student Attendance Management System
- **UI** – User Interface
- **API** – Application Programming Interface

2. Overall description

2.1 Product perspective: -

The Student Attendance Management System (SAMS) is a application that interacts with university authentication services, biometric devices, and institutional databases. It provides students with secure access to their attendance records and enables faculty to mark and update attendance digitally. The system includes:

- Mobile and web applications for students and faculty
- Authentication services (biometric and institutional login)
- Attendance database for storage and reporting

2.2 Major product functions (detailed)-

User Authentication

- Login via institutional ID and password, with biometric verification (fingerprint/FaceID) where supported.
- Automatic role detection (student, faculty, or admin) based on login credentials.

Attendance Marking (Faculty)

- Mark attendance in real-time during class sessions.
- Update or correct attendance entries, with mandatory justification.
- Export/download attendance reports (per class, per subject, per month/semester).

Attendance Tracking (Student)

- View overall attendance percentage for all enrolled subjects.
- Drill down into subject-wise attendance for details.
- Access a day-wise record showing presence/absence for each class session.

- No automated notifications for attendance below threshold (e.g., 75%) — students are responsible for self-monitoring.

2.3 User roles and characteristics (expanded)-

Student: Limited technical background, expects quick access to attendance percentage and clear daily records (present/absent).

Faculty: Time-constrained during classes, requires a simple interface to mark or edit attendance and download reports.

Administrator: Responsible for monitoring compliance, auditing edits, and generating institutional-level reports.

2.4 Operating environment Embedded:

Mobile app (Android 10+ and iOS 14+) and responsive web interface.

Secure backend servers hosted on cloud or institutional infrastructure.

Relational database for attendance storage.

Biometric authentication hardware or mobile device sensors (fingerprint).

2.5 ConstraintsEMV compliance:

Must integrate only with institution-approved biometric authentication hardware.

Application must run on both mobile data and campus Wi-Fi networks.

Attendance data entry restricted to faculty accounts; students cannot modify records.

Class schedules and student lists must sync daily from the central academic database.

3. External interface requirements

3.1 User Interfaces

- **Mobile App**: Students and faculty can log in and use the app on Android.

3.2 Hardware Interfaces

- **Faculty Mobile / Laptop** – Faculty can mark attendance manually through the app or web portal (no extra hardware needed).

3.3 Software Interfaces

- **Backend APIs**: REST APIs connect the app with the database.

3.4 Communications

- **Secure Data Transfer**: All communication uses TLS 1.2+ encryption.
- **Real-Time Sync**: Attendance data syncs instantly, with retry if the network fails.

<< Make sure overall there are at least 15 FRs for overall project, 5 NFRs, 2 security objectives and 5 Security requirements>>

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow SAMS-F-###.

4.1 Authentication

Description: Authenticate the cardholder using EMV card and PIN; support chip & contactless where applicable.

Req ID	Requirement	Type	Priority	Source/Stake holder	Acceptance criteria / Test case ref	Comments / Dependencies
SAMS-F-001	The system shall authenticate user by biometric(finger print) or the password	Functional	High	Security / University IT	AC-SAMS-F-001: User login succeeds only with valid credentials. Test: TC-Auth-01	Fingerprint (Biometric), password, Integration
SAMS-F-002	The system shall enforce role-based access (Student / Faculty).	Functional	High	Faculty	AC-SAMS-F-002: Users can access only features allowed by their role. Test: TC-Auth-02	Role definitions must be pre-configured
SAMS-F-003	The system shall lock accounts after 5 failed login attempts.	Functional	High	Security, IT Policy	AC-SAMS-F-003: After 5 failed attempts, login is blocked and notification sent. Test: TC-Auth-03	Requires account status update.

4.2 Attendance Management

Description: This feature enables faculty to record and update student attendance digitally, while allowing students to view real-time attendance percentages and detailed daily records.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SAMS-F-010	Faculty shall be able to mark attendance in real time via mobile and web apps.	Functional	High	Faculty	AC-F-010: Attendance saved instantly after marking. Test: TC-Att-01	Requires stable network connection
SAMS-F-011	Students shall be able to check attendance percentage and subject-wise breakdown.	Functional	High	Student	AC-F-011: Student view shows correct percentage per subject. Test: TC-Att-02	Dependent on updated database
SAMS-F-012	Faculty shall be able to edit attendance with justification, with audit logging.	Functional	High	Faculty/Admin	AC-F-012: Edit saved only with justification logged. Test: TC-Att-05AC-F-014: Edit saved only with justification logged. Test: TC-Att-03	
SAMS-F-013	The system shall display daily presence/absence records to students, updated in real time.	Functional	High	Student	AC-F-013: Day-wise status shown correctly. Test: TC-Att-04	Depends on timely faculty submission

4.3 Reporting & Analytics

Description :The system provides real-time reports that summarize student attendance at course and semester levels. It enables faculty and students to track attendance trends and identify defaulters effectively.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SAMS-F-020	The system shall allow students to view **course-wise and overall attendance percentage** through the web/mobile app.	Functional	High	Students	AC-SAMS-F-020: Student logs in → sees percentage per course and overall. Test: TC-Rep-01	Data pulled from attendance DB
SAMS-F-021	The system shall allow faculty to **download attendance reports** (daily/weekly) in simple table formats (CSV/PDF).	Functional	High	Faculty	AC-SAMS-F-021: Faculty selects course/date → system generates downloadable file. Test: TC-Rep-02	Requires basic export module
SAMS-F-022	The system shall allow administrators to **generate defaulter lists** for students below a set threshold (e.g., <75%).	Functional	High	Admin	AC-SAMS-F-022: Admin sets threshold → system outputs list of low-attendance students. Test: TC-Rep-03	Threshold configurable
SAMS-F-023	The system shall provide a **dashboard** with simple attendance trends (e.g., per-course averages, overall percentage charts).	Functional	Medium	Admin/Faculty	AC-SAMS-F-023: Dashboard loads with bar/pie chart and percentage values. Test: TC-Rep-04	Needs charting library
SAMS-F-024	The system shall allow faculty to **filter attendance reports by course or date range**.	Functional	Medium	Faculty	AC-SAMS-F-024: Faculty applies filters → filtered report generated. Test: TC-Rep-05	Basic DB query filters
SAMS-F-025	The system shall send **alerts to admins** if attendance in any course falls below a critical	Functional	Medium	Admin	AC-SAMS-F-025: Attendance drops < threshold → system sends alert/notificati	Needs simple notification svc

	threshold (e.g., sudden mass				on. Test: TC-Rep-06	

4.4 Maintenance & Monitoring

Description: This feature provides administrative oversight of system operations, ensuring integrity of attendance records and secure usage through logging, monitoring, and alerts.

Req ID	Requirement (shall…)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAMS-F-030	The system shall provide audit logs for all attendance edits, including timestamp, faculty ID, old value, new value, and justification.	Functional	High	Admin / Compliance	AC-F-030: Every edit is logged with full details. Test: TC-Mon-01	Logs must be immutable and stored securely.
SAMS-F-031	The system shall monitor login activities (failed logins, multiple logins from different locations) and generate alerts for suspicious activity.	Functional	High	Security / Admin	AC-F-031: Alert triggered after ≥5 failed attempts or geo-location anomaly. Test: TC-Mon-02	Alerts sent to system admin dashboard.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs ATM-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SAMS-NF-001	The system shall respond to of user requests within few minutes.	Performance	High	Load testing confirms response time within few minutes for standard transactions.
SAMS-NF-002	The system shall provide continuous availability during academic semesters.	Reliability	High	Monitoring reports.
SAMS-NF-003	Attendance records shall be retained securely for at least 5 years.	Data Retention	High	Database archival and retrieval verified through test cases.
SAMS-NF-004	The mobile app shall support Android.	Portability	Medium	QA testing on Android devices confirms successful operation.
SAMS-NF-005	The system shall be able to handle at least 1000 concurrent users during peak usage.	Scalability	High	Load and stress tests confirm stability at the required user load.

5.1. Security

5.1.1 Security Objectives

- Protect student attendance data against unauthorized access.
- Ensure secure communication between clients and server.

5.1.2 Security Requirements

ID	Requirement (shall…)	Type	Priority	Acceptance/Test Case Ref
SAMS-SR-001	The system shall enforce TLS 1.2+ for all client-server communication.	Security	High	TC-SR-001
SAMS-SR-002	The system shall encrypt data at rest using AES-256.	Security	High	TC-SR-002
SAMS-SR-003	The system shall implement role-based access to restrict sensitive edits.	Security	High	TC-SR-003
SAMS-SR-004	The system shall maintain tamper-proof audit logs.	Security	Medium	TC-SR-004
SAMS-SR-005	The system shall enforce session timeout after 15 minutes inactivity.	Security	Medium	TC-SR-005

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance

Functional Completion

- All functional requirements (attendance marking, student view) are implemented and tested successfully.
- Faculty can mark attendance for classes without errors.
- Students can view subject-wise attendance records and percentages in real-time.

Non-Functional Compliance

- The system meets performance benchmarks.
- Mobile and web apps run without crashes for most of the test cases.

Data Integrity

- Attendance data is stored securely and matches the records in the database.
- No data loss or duplication occurs after 1,000 consecutive transactions.

Security & Authentication

- Unauthorized users cannot access data.

Stability & Reliability

- The app runs continuously for at least 7 days without unhandled crashes.

All high-priority functional requirements (Login, Mark Attendance, View Attendance, Manual Attendance by Teacher, Reports) are implemented and verified.

No critical non-functional failures (performance, usability, security).

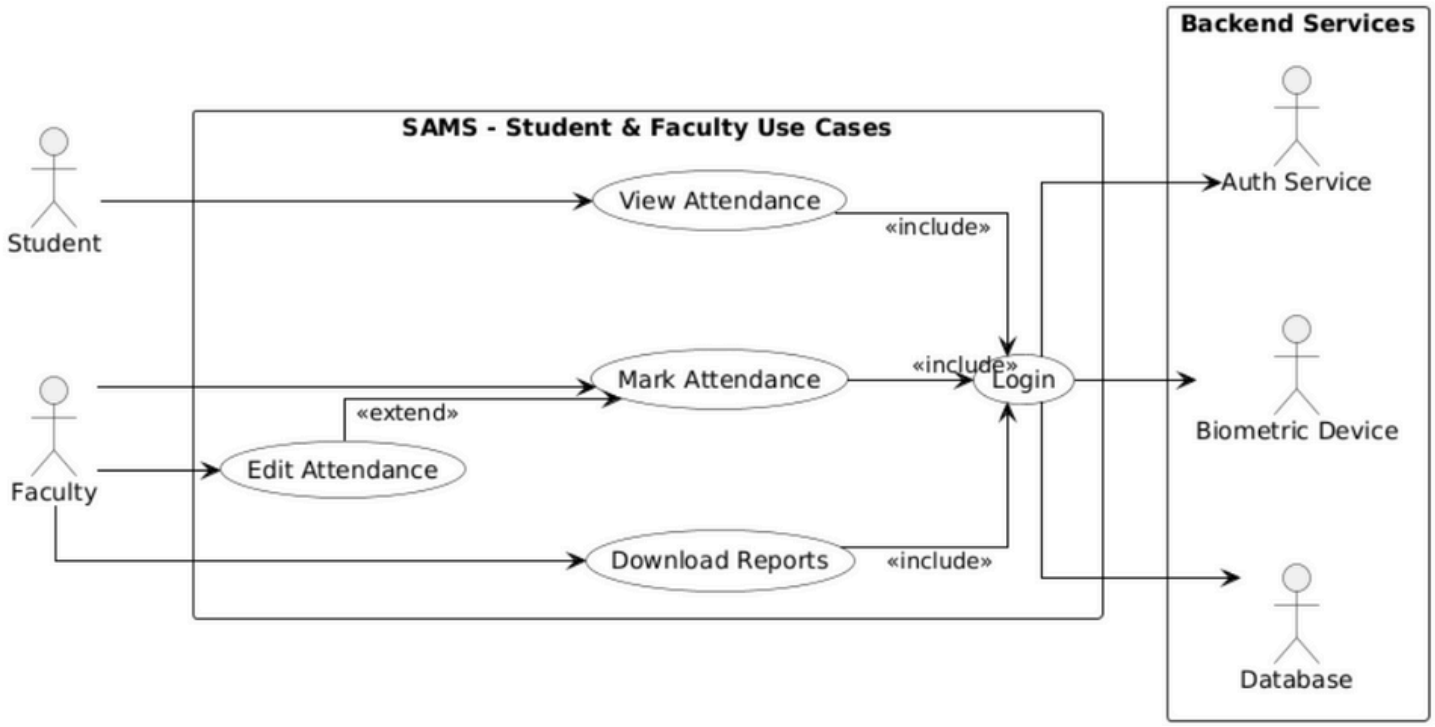
- Acceptance test suites

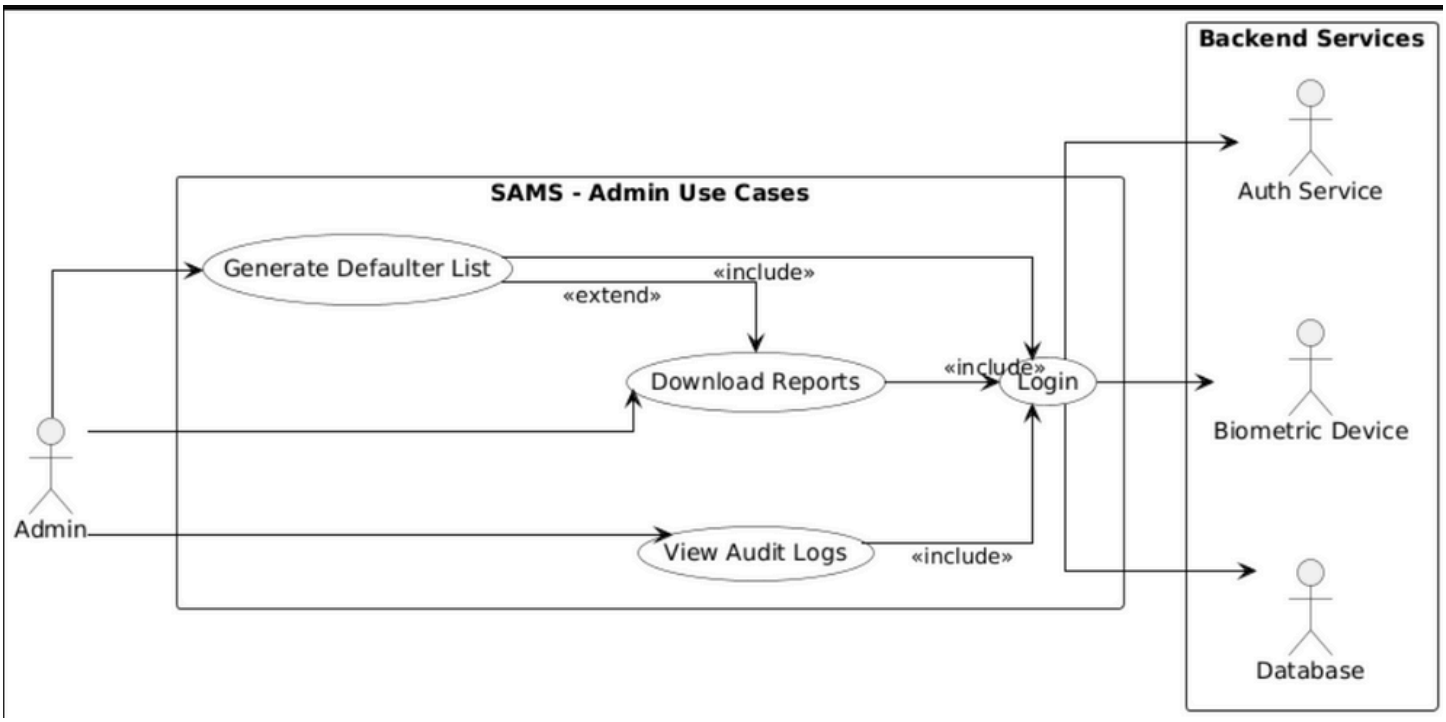
- **Authentication Tests** → Verify student/teacher login with valid/invalid credentials.
- **Attendance Marking Tests** → Check QR/manual marking works and prevents duplicate entries.
- **Manual Attendance Tests** → Teacher can mark/edit attendance in online mode.
- **Performance Tests** → App loads in few minutes and supports 100+ users at once.
- **Security Tests** → Data is stored securely, unauthorized access is blocked.
- **Accessibility Tests** → App usable on mobile (Android), easy UI for teachers/students.

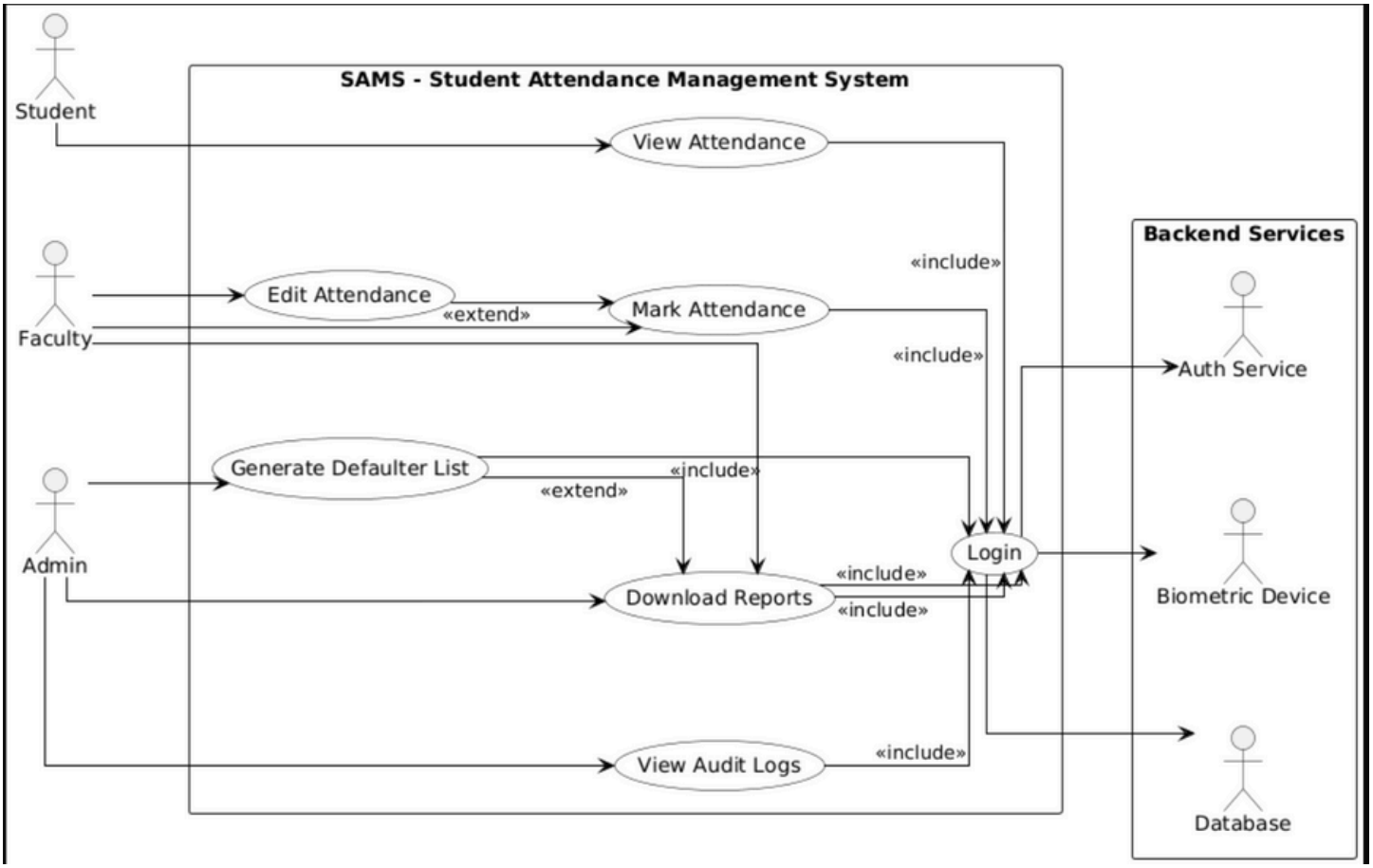
7. System models and diagrams

7.1 UML Use-Case diagram

At least 2 UML use case diagrams to be created.




Backend Services
Auth Service
SAMS - Admin Use Cases
Generate Defaulter List
«include»
«extend»
Download Reports
«include»
Login
Biometric Device
Admin
View Audit Logs
«include»
Database


Student
SAMS - Student Attendance Management System
View Attendance
Backend Services
«include»
Edit Attendance
«extend»
Mark Attendance
Faculty
«include»
Auth Service
Generate Defaulter List
«include»
«extend»
Login
Admin
«include»
Download Reports
«include»
Biometric Device
View Audit Logs
«include»
Database

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SAMS-F-001	User Authentication(password/biometric)	4.1 / DS-Auth-01	Auth Module	TC-Auth-01	N	
SAMS-F-010	Faculty marks attendance	4.2 / DS-Att-01	Attendance Module	TC-Att-01	N	
SAMS-F-020	Student views sub-wise attendance	4.3/DS-Rep-01	Report Module	TC-Rep-01	N	
SAMS-F-030	Audit logs for all edits	4.4/DS-Log-01	Log Module	TC-Log-01	N	
SAMS-NF-001	Response time target	5 / DS-Perf-01	WebUI / CoreAPI	TC-Perf-01	N	